

Minjae Seo

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EDUCATION

University of Chicago, Harris School of Public Policy

Master of Arts in Public Policy, Certificate in Research Methods

Selected Coursework: Advanced Microeconomics for Policy Analysis I–II (PhD), Applied Econometrics I–III (PhD), Game Theory (PhD), Analysis in R^n , Research Methodology and Writing

June 2027

GPA: 3.7/4.0

University of California, Berkeley

Bachelor of Arts in Statistics

Selected Coursework: Modern Statistical Prediction & Machine Learning, Principles & Techniques of Data Science, Econometrics, Causal Inference, Optimization Models in Engineering, Game Theory

Dec 2024

GPA: 3.85/4.0

SKILLS

Programming: Python (fluent), SQL, R, Stata, C, MATLAB, SAS

Data Engineering & Cloud: chunked SQL/ETL, AWS, WRDS Cloud, pandas/NumPy, Git, Tableau, Power BI

ML & Methods: scikit-learn, ML prediction, causal inference (DiD, RDD, IV, synthetic control), panel/fixed-effects, partial identification

INDUSTRY EXPERIENCE

Research Analyst Intern, World Bank, Washington, DC

Jun 2026 – Aug 2026

- Synthesize literature and evidence on tax compliance and behavioral policy to inform development-project design and recommendations for raising tax revenue in developing countries.
- Analyze qualitative and quantitative data for evidence-based interventions via econometric and machine-learning methods; build data visualizations and policy briefs for internal and external stakeholders.

AI Intern, YCX, Remote

Mar 2025 – May 2025

- Built and deployed an AI-driven Know-Your-Business (KYB) compliance system—multi-agent orchestration, LLM integration (Groq API), targeted web scraping—automating regulatory due diligence in production.
- Extended firm-level coverage to 300+ entities (a 40% increase) through a modular agent pipeline that combined open-web sources, corporate filings, and JSON-formatted LLM output.

Quantitative Research Intern, Global Key Advisors, Hybrid

Jul 2024 – Mar 2025

- Merged SEC Form 4, CRSP, and Refinitiv data in Python across the Fama–French 48 industries; built reusable extraction and ROI-tracking pipelines that fed the team’s downstream strategy backtests.
- Constructed event-time abnormal-return signals around insider-trading filings and backtested industry-level factor portfolios to test whether disclosure rules left exploitable alpha, benchmarking risk-adjusted performance against Fama–French factors.

RESEARCH EXPERIENCE

Working Papers, University of Chicago

Jan 2026 – Present

“Homogeneous and Heterogeneous Treatment Effects in the Presence of Spillovers: A Simulation-Based Approach in Health and Education Sectors”

- Derived sharp partial-identification bounds on the average (homogeneous) treatment effect when interference violates SUTVA, applying Fréchet–Hoeffding inequalities to the joint distribution of potential outcomes; examined effect heterogeneity across covariate-defined subgroups.
- Implemented Monte Carlo simulations to assess finite-sample bound width and tightness across spillover intensities and multi-treatment regimes, calibrated to health- and education-sector interventions where peer effects are first-order.

“AI Exposure, Firm Size, and Workforce Restructuring: A Triple-Difference Approach Using Revelio Labs Data”

- Designed a triple-difference (DDD) strategy testing whether AI-exposed workers face higher separation at large firms after the Nov 2022 ChatGPT release; firm-time fixed effects absorb firm-wide shocks, identifying off within-firm AI-exposure asymmetry.
- Linked Revelio Labs worker data (700M+ profiles, 2.4B U.S. job postings) to Compustat financials and two AI-exposure measures (Felten AIOE, Eloundou GPT) via an rcid–ticker–gykey crosswalk; decomposed outcomes into outflow, inflow, net change, and churn.

Research Assistant, Booth School of Business, University of Chicago

Jan 2026 – Present

Supervisor: Prof. Anthony Le

- Built a Stata pipeline processing 899 Excel files (448,119 citation rows) from the Accountants’ Index (1920–2000) into a ranked database of 14,500 journals with fuzzy name normalization and three-pass QC, cross-validated against a parallel Python build.
- Extracted the Revelio COSMOS U.S. job-postings dataset (2.38B rows, 4.90 TiB) on WRDS Cloud using chunked state-by-year SQL queries with automated resume.

Research Assistant, University of Chicago

Sep 2025 – May 2026

Channels of Negative Earnings for Community-College Transfer Students

- Estimated the wage gap between two-to-four-year transfer and direct-entry BA holders (NLSY97, $N = 2,306$) with OLS and inverse-propensity weighting; the IPW estimate of $-\$7,770/\text{yr}$ ($p = 0.048$) matches the $-\$7,000$ administrative benchmark in Miller (2026).
- Showed a previously reported non-cognitive control effect was a sample-composition artifact: holding the sample fixed ($N = 740$) cut the Achenbach attenuation from 27.5% to 4.5%.

Research Assistant, UC Berkeley

Jan 2024 – Feb 2025

Causal Effect of Central Heating on Indoor Temperature (Huai River, China)

- Assembled a 32-city north/south panel from the Chinese Thermal Comfort Dataset for a sharp RD at the 33°N heating-policy boundary (`rdrobust`, `R`); estimated a 4.3°C rise in indoor temperature via WLS with city fixed effects and polynomial trends.

TEACHING

Graduate Teaching Assistant, Mathematical Foundations of Machine Learning, University of Chicago

Sep 2025 – Dec 2025

- Taught weekly recitations for 20+ master’s students with Python/NumPy demos on SVD, Gram–Schmidt, and least squares; resolved 50+ technical questions on Ed Discussion.

LEADERSHIP

Head of Research, Undergraduate Economics Association, UC Berkeley

Jan 2024 – Mar 2025

- Led biweekly research workshops for 30+ undergraduates, training members in core econometric methods (OLS, IV, DiD, RDD) and R/Stata; mentored semester-long projects on identification, data work, and write-up.

Competitor, Econometrics Game, University of Chicago

Mar 2024

- Selected as one of four (with two LSE and one UC Berkeley student) for a 48-hour empirical research competition; presented findings to a panel including a Nobel Laureate in Economics.